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EXPLORING OPTIONS AND CLUSTER EMERGENCE

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1 Introduction

Motivation

The theory of *cluster growth* on random graphs, epitomised by Durrett’s textbook treatment of breadth-first search (BFS) on $G(n, \lambda/n)$ [Dur10], and the theory of *exploration* in reinforcement learning (RL), recently advanced by Jinnai’s *Covering Options* framework [JPAK19, JPMK20], have been developed almost entirely in parallel. Both lines of work analyze how a simple random walk spreads through an unknown combinatorial structure, and both pinpoint a single spectral parameter that governs a *phase-transition* from slow to fast coverage. Yet the two communities have used different languages: Durrett speaks of Galton-Watson drift and giant components, while Jinnai speaks of Laplacian gaps and cover times. Bridging these vocabularies yields a unified view of *why* and *how* long-range shortcuts—edges in a graph or options in an MDP—sharply accelerate exploration.

Gap in the Literature

Existing RL analyses typically start from the Broder–Karlin cover-time bound [BK89]; random-graph analyses instead rely on branching-process couplings. The correspondence between the two large-deviation exponents has not, to our knowledge, been made explicit. Consequently, RL practitioners lack a percolation-style intuition for when exploration fails catastrophically, while random-graph theorists have not exploited option-based shortcuts such as those introduced by Ghosh and Boyd’s algebraic-connectivity edge additions [GB06].

In this project, we aim to close that gap by showing that Durrett’s drift term $\lambda(1 - \delta) - 1$ and the Laplacian spectral gap $\lambda_2(L)/(2n^2)$ are *mathematically identical* control parameters for the extinction probability of a BFS frontier and the tail of an RL cover time. Building on this identification we:

1. derive an explicit mapping (Section 4) between Durrett’s extinction bound and Broder–Karlin’s cover-time inequality;
2. prove that Jinnai’s point-options are the minimal-energy perturbations that drive the exploration process across the same critical threshold that produces a giant component in random graphs;
3. provide a practical algorithmic recipe that guarantees $O(n \log n)$ discovery time for sparse rewards in tabular MDPs, mirroring the $O(n \log n)$ time to expose the giant component in the supercritical Erdős–Rényi regime.

By importing percolation intuition into RL, we obtain sharper failure diagnoses and principled shortcut designs; by exporting option-based shortcuts to random-graph theory, we suggest new controlled edge-addition experiments beyond uniform sprinkling. We believe this cross-fertilises two mature but until now largely disjoint research programmes.

2 Covering Options

We consider a very basic question in reinforcement learning: how to explore a given state space S efficiently. Let $G = G(S, E)$ be the state-transition graph. Early in learning,

an agent behaves roughly like a uniform random walk on G . We let:

$$C(G) = \max_{s \in S} \max_{j \in S} \mathbb{E}[H_{s \rightarrow j}]$$

be the expected cover time—the time a random walk needs to visit every state. Theorem 1 in Jinnai shows that for any stochastic shortest-path problem with step cost $r_c \leq 0$ and $\gamma = 1$,

$$V_\pi(s) \geq r_c \mathbb{E}[C(G)], \quad \forall s \in S,$$

so lowering $C(G)$ strictly lowers a worst-case bound on exploration time. We then clearly have a spectral upper bound. Broder & Karlin’s bound couples cover time of the algebraic connectivity $\lambda_2(L)$, i.e., the second-smallest eigenvalue of the normalised Laplacian L :

$$\mathbb{E}[C(G)] \leq \frac{n^2 \ln n}{\lambda_2(L)} (1 + o(1))$$

where $n = |S|$. Therefore, raising λ_2 provably accelerates exploration as Jinnai observes. We can then add a point option (initiation and termination are single states), which is equivalent to inserting a new edge. Ghosh & Boyd show that joining the two nodes with maximal and minimal coordinates in the Fiedler vector f maximally increases λ_2 to first order, which gives us the “Covering Options” algorithm:

1. Compute f of L .
2. Let $v_{\max} = \arg \max f$ and $v_{\min} = \arg \min f$.
3. Add two point-options ($v_{\max} \rightarrow v_{\min}$) and the reverse.

Each option monotonically tightens the above bound. Empirically, a handful of such options cut learning time by 1.5 to 4 times in sparse-reward problem domains, such as mazes. We then want to lift the cover-time theory to continuous or very large state spaces.

2.1 Spectral Regression

Instead of the exact Laplacian eigenvector, we can learn a parameterised eigenfunction ϕ_θ that minimises:

$$\tilde{G}(\phi) = \frac{1}{2} \sum_{(s,s') \in \mathcal{H}} w(s,s') (\phi(s) - \phi(s'))^2$$

over a replay buffer $\mathcal{H}$ of transitions. This is an unconstrained least-squares problem that can be optimized with SGD. The resulting ϕ_θ approximates the Fiedler vector. We can prove this. We let:

- $G = (S, E, w)$ be an undirected weighted graph of the state-space (each transition (s, s') induces an undirected edge with weight $w_{ss'}$)
- $D = \text{diag}(d_s)$ with $d_s = \sum_{s'} w_{ss'}$ the degree matrix
- $L = D - W$ the unnormalised graph Laplacian

The Fiedler vector f^* is the solution of the constrained minimisation:

$$\lambda_2 = \min_{f^\top \mathbf{1}, f^\top Df=1} \frac{1}{2} \sum_{(s,s') \in E} w_{ss'} (f(s) - f(s'))^2 = \min_{f^\top \mathbf{1}, f^\top Df=1} f^\top Lf$$

which is the Rayleigh-quotient variational form of the second eigenvalue. The two constraints eliminate the trivial constant eigen-function and fix the scale. We define:

$$\mathcal{E}(f) = \frac{1}{2} \sum_{(s,s')} w_{ss'} (f(s) - f(s'))^2$$

where direct expansion gives:

$$\mathcal{E}(f) = \frac{1}{2} (f^\top Df - 2f^\top Wf + f^\top Df) = f^\top (D - W)f = f^\top Lf$$

which means minimizing $\mathcal{E}$ under the constraints is equivalent to solving the eigen problem $Lf = \lambda Df$. The first-order stationarity condition of λ_2 with Langrage multipliers α, β is:

$$Lf = \alpha Df + \beta D\mathbf{1} \Rightarrow Lf = \lambda Df, \quad (\lambda = \alpha, \beta = 0, f^\top \mathbf{1})$$

In large or continuous S we do not have L explicitly. Jinnai instead collects a buffer $H = \{(s_i, s'_i)\}_{i=1}^m$ of transitions drawn i.i.d. from a behaviour policy. We can then write:

$$\hat{\mathcal{E}}_H(f) = \frac{1}{2m} \sum_{(s,s') \in H} (f(s) - f(s'))^2.$$

We have:

$$\mathbb{E}_H[\hat{\mathcal{E}}_H(f)] = \frac{1}{2\mu(E)} \sum_{(s,s')} w_{ss'} (f(s) - f(s'))^2 \propto \mathcal{E}(f)$$

where $\mu(E) = \sum_{(s,s')} w_{ss'}$. By the law of large numbers, the sample objective converges uniformly to the population quadratic form as $|H| \rightarrow \infty$; thus any minimiser of the empirical objective is consistent for the true Fiedler vector. Wu then shows that one can **ENFORCE** both unit-norm and orthogonality via a repulsive penalty instead of hard constraints. Specialising their multi-eigenfunction loss to one non-trivial eigenfunction gives Jinnai eq.5:

$$\tilde{G}_H(f) = \underbrace{\frac{1}{2} \mathbb{E}_{(s,s') \sim H} [(f(s) - f(s'))^2]}_{\hat{\mathcal{E}}_H(f)} + \underbrace{\eta \mathbb{E}_{s \sim \rho} [(f(s)^2 - 1)]^2}_{\text{scale+mean penalty}}$$

where the first term is the empirical Laplacian quadratic form and the second pushes f towards zero mean ($f^\top \mathbf{1}$) and unit variance ($f^\top Df = 1$). Minimising a perfect square drives the constraint violation to zero as $\eta \rightarrow \infty$. Which means at any finite but large η , any stationary point of $\tilde{G}_H(f)$ satisfies:

$$\nabla_f \hat{\mathcal{E}}_H(f) = 2\eta \nabla_f [(f^\top Df - 1) + (f^\top D\mathbf{1})^2] = 0$$

which in the limit $\eta \rightarrow \infty$ yeilds the KKT system of λ_2 :

$$Lf = \lambda Df, \quad f^\top Df = 1, \quad f^\top D\mathbf{1} = 0$$

because $\lambda_f \hat{\mathcal{E}}_H(f) = Lf$ (component-wise: $\frac{\partial \mathcal{E}}{\partial f(s)} = \sum_{s'} w_{ss'}(f(s) - f(s'))$) gradient descent on $\tilde{G}_H(f)$ therefore converges to the empirical Fiedler vector; by consistency of $\hat{\mathcal{E}}_H$ it converges in probability to the true Fiedler eigen-function as $|H| \rightarrow \infty$. We can then show clearly from this derivation why:

$$\tilde{G}(f) = \frac{1}{2} \mathbb{E}_{(s,s') \sim H} [(f(s) - f(s'))^2] + \eta \mathbb{E}_{\substack{s \sim \rho \\ s' \sim \rho}} [(f(s)^2 - 1)(f(s') - 1) + f(s)^2 f(s')^2]$$

works in Jinnai. First, we look at the energy equivalence. The first time of $\tilde{G}_H$ is the Laplacian quadratic form. We then see the second term is zero iff the constant-function and unit-form constraints hold; with a sufficiently large η , the minimiser is arbitrarily close to satisfying them. We then look at the stationary condition where the gradient sets $Lf = \lambda Df$, so any local minimiser is an eigenfunction. We also look at eigen-ordering where the constant eigen-function already drives $\hat{\mathcal{E}}_H$ to its global minimum (energy 0), which tells us that the next lowest-energy critical point produced by SGD is precisely the second-smallest eigen-value/vector pair, i.e., the Fiedler vector. Finally, as $m \rightarrow \infty$, $\hat{\mathcal{E}}_H \xrightarrow{\text{u.c.}} \mathcal{E}$. Standard perturbation results for symmetric matrices, therefore, guarantee that the learned eigenfunction converges to the population Fiedler function. Thus minimising $\tilde{G}(f)$ is scalable surrogate for solving λ_2 allowing Deep Covering Options to approximate the algebraic connectivity of enormous or continuous state graphs with stochastic gradient descent on raw trajectories while preserving the theoretical link between λ_2 and the cover-time bound, concluding the proof. From this, we can then pick a percentile k of ϕ_θ . States below that threshold become the termination set of a new option, which complements its initiation set. Train the option's policy off-policy with pseudo-rewards $\Delta\phi = \phi(s) - \phi(s')$. Repeating yields a small option library that steers the agent into under-explored regions and empirically lifts sample efficiency on Pinball, MuJoCO, and Atari tasks. Clearly $\mathbb{E}[C(G)]$ still governs the bound, and ϕ_θ is simply a scalable proxy for the exact eigenvector.

2.2 Minimizing the Planning Time

While Covering Options targets exploration, this companion paper asks: *Which options make value-iteration converge fastest?* We then have the following two concepts:

2.2.1 Complexity Results

Theorem 2: Given an MDP and an iteration budget ℓ , choosing the smallest set of options that guarantees convergence within ℓ VI sweeps is **NP-hard** and $\Omega(\log n)$ -hard to approximate (even in deterministic MDPs).

2.2.2 Approximation Algorithms

- **A-MOMI** (minimum number of options for a time budget) - an $\mathcal{O}(n)$ -approximation in general, $\mathcal{O}(\log n)$ in deterministic MDPs
- **A-MIMO** (k-option budget $\Rightarrow$ minimise VI iterations)

Both greedily choose options that maximize Bellman residual reduction, giving the first probable guarantees for option-based planning speed-ups.

2.3 Lipschitz Lifelong Reinforcement Learning

2.3.1 Problem

We can now approach the problem of Lipschitz Lifelong RL introduced by Jinnai. An agent faces a sequence of finite, discounted MDPs $\mathcal{M}(\mathcal{S}, \mathcal{A}, R, T, \gamma)$ drawn from an unknown distribution. We want to reuse knowledge from earlier tasks without sacrificing the Probably-Approximately-Correct (PAC) guarantees of a tabular learner such as **R-Max**.

2.3.2 Task-space pseudometric

We let $Q_{\mathcal{M}}^{\pi}$ be the value of policy π in task $\mathcal{M}$. We then define the distance between two tasks $\mathcal{M} = (R, T)$ and $\mathcal{M}' = (R', T')$ as the least fixed-point of:

$$d((R, T), (R', T')) = \max_{s,a} \left(|R(s, a) - R'(s, a)| + \gamma \sum_{s'} T(s'|s, a) d_{s',a}^{\text{succ}} \right)$$

where $d_{s',a}^{\text{succ}}$ is the same metric evaluated on successor pairs, which is exactly Eq.3 in Jinnai.

2.3.3 Lipschitz Continuity of Optimal Values (Prop 1)

We have nearby tasks that have near-identical optimal action-values as implied by below:

$$|Q_{\mathcal{M}}^*(s, a) - Q_{\mathcal{M}'}^*(s, a)| \leq L d((R, T), (R', T')), \quad L = (1 - \gamma)^{-1}$$

2.3.4 LRMax Algorithm

We start with the R-Max optimism principle, but replace the generic bonus:

$$\beta_t(s, a) = \mathcal{O}\left((1 - \gamma)^{-1} \sqrt{\frac{\log(1/\delta)}{N_t(s, a)}}\right).$$

With the task-dependent bound supplied by the Lipschitz inequality above. The resulting Lipschitz-RMax maintains the same PAC-MDP sample-complexity bound:

$$\mathcal{O}(|\mathcal{S}||\mathcal{A}| \text{poly}((1 - \gamma)^{-1}, \epsilon^{-1}, \log(1/\delta))).$$

But the constants are sharply reduced whenever the current task lies close to a previously solved one.

2.4 Policy and Value Transfer for Lifelong RL

We take the following scenario. Tasks $\mathcal{M}_1, \mathcal{M}_2, \dots$ arrive sequentially, i.i.d. from an unknown distribution $\mathcal{D}$.

2.4.1 Bayes-Optimal Warm-Start Policy

Let $\mathcal{D}$ denote the (unknown) distribution over finite MDPs $\mathcal{M} = (\mathcal{S}, \mathcal{A}, R, T, \gamma)$. Upon arrival of a new task $\mathcal{M}_k \sim \mathcal{D}$, the agent may *warm-start* with a policy that is Bayes-optimal under $\mathcal{D}$, i.e. it maximises the expected return *before* any new data are collected.

Deterministic policies. For the class of stationary deterministic policies, the Bayes-optimal choice is

$$\pi^\dagger(s) = \arg \max_{a \in \mathcal{A}} \mathbb{E}_{\mathcal{M} \sim \mathcal{D}} [Q_{\mathcal{M}}^*(s, a)], \quad \forall s \in \mathcal{S}, \quad (1)$$

where $Q_{\mathcal{M}}^*$ is the optimal state-action value function in task $\mathcal{M}$.

Stochastic policies. An analogous expression holds for stationary stochastic policies:

$$\pi_{\text{stoch}}^\dagger(\cdot | s) = \arg \max_{\pi(\cdot | s)} \mathbb{E}_{\mathcal{M} \sim \mathcal{D}} \left[\sum_{a \in \mathcal{A}} \pi(a | s) Q_{\mathcal{M}}^*(s, a) \right], \quad (2)$$

with the maximisation taken over all distributions on $\mathcal{A}$.

Belief-space policies. If a posterior belief μ_k over $\mathcal{D}$ is maintained after observing data from prior tasks, the Bayes-optimal *belief-state* policy is obtained by replacing the outer expectation in [Equation 1](#)-[Equation 2](#) with $\mathbb{E}_{\mathcal{M} \sim \mu_k}[\cdot]$.

Because the expectation is taken with respect to the same distribution that generates future tasks, choosing $\pi^\dagger$ maximises the agent's *initial* expected return on $\mathcal{M}_k$ before any on-task learning occurs.

2.4.2 MAXQInit: Value-Function Transfer

Assume a library $\{Q_*^{(i)}\}_{i \in \mathcal{I}}$ of optimal state-action value functions from previously solved tasks $\{\mathcal{M}_i\}$. Before interacting with a new task $\mathcal{M}_k$, initialise the value function by taking the *pointwise maximum* over the library:

$$Q_0(s, a) = \max_{i \in \mathcal{I}} Q_*^{(i)}(s, a), \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A}. \quad (3)$$

Theoretical guarantee. Each $Q_*^{(i)}$ is optimistic for an unseen task, hence [Equation 3](#) preserves the optimism condition required by PAC-MDP algorithms such as R-MAX and Delayed Q-LEARNING. Consequently, MAXQInit retains the original sample-complexity bound $\tilde{O}(|\mathcal{S}| |\mathcal{A}| (1 - \gamma)^{-3} \epsilon^{-2})$, while empirically reducing the number of state-action pairs that must be explored in early episodes.

2.5 Unified Perspective on Jinnai’s Contributions

Table 1: Structural bottlenecks addressed by Jinnai’s work and the associated theoretical tools

Research goal	Efficient exploration
Quantity controlled	Cover time $C(G)$ of the state graph
Principal result	COVERING OPTIONS raise the algebraic connectivity $\lambda_2(L)$, yielding $\mathbb{E}[C(G)] \leq n^2 \log n / \lambda_2(L)$
Analytical toolset	Spectral graph theory; Broder-Karlin cover-time bound
Research goal	Fast planning
Quantity controlled	No. value-iteration sweeps to ε -optimality
Principal result	Option-selection problem is NP-hard; A-MIMO/A-MOMI give $O(\log n)$ approximations
Analytical toolset	Set-Cover reductions; Bellman-residual analysis
Research goal	Safe lifelong transfer
Quantity controlled	Task-space distance $d((R, T), (R', T'))$
Principal result	LIPSCHITZ-RMAX shrinks optimism bonuses while keeping PAC guarantees
Analytical toolset	Task pseudometric; Lipschitz continuity; PAC-MDP theory
Research goal	Rapid warm-starts
Quantity controlled	Initial value / policy quality
Principal result	Bayes-optimal warm-start policies and MAXQInit cut early exploration
Analytical toolset	Bayesian decision theory; PAC transfer proofs

2.6 Connecting the Thread — One Playbook, Many Problems

Jinnai’s papers, though addressing different aspects of reinforcement learning, follow a *single three-step blueprint*:

1. Diagnose the bottleneck. Pinpoint the quantity that directly limits progress:

- *Exploration*: cover time $C(G)$ of the state graph.
- *Planning*: number of value-iteration (VI) sweeps.
- *Transfer*: magnitude of the optimism bonus.

2. Convert the bottleneck into a bound with a tunable knob. Derive an explicit inequality whose right-hand side contains a parameter that the agent can influence:

$$\mathbb{E}[C(G)] \leq \frac{n^2 \log n}{\lambda_2(L)} \quad (\text{knob: algebraic connectivity } \lambda_2(L))$$

$$\text{VI sweeps} \leq O(\log n) \times |\mathcal{O}| \quad (\text{knob: option set size } |\mathcal{O}|)$$

$$\text{Optimism bonus} \leq Ld((R, T), (R', T')). \quad (\text{knob: Lipschitz constant } L)$$

3. Intervene to tighten the bound while keeping PAC guarantees. Design skills or transfer rules that *push the knob* in the favourable direction:

- Add *covering options* $\Rightarrow$ increase $\lambda_2(L) \Rightarrow$ shrink $C(G)$.
- Choose options via A-MIMO/A-MOMI $\Rightarrow$ cut required VI sweeps.
- Apply LIPSCHITZ-RMAX or MAXQINIT $\Rightarrow$ lower optimism bonuses.

In essence, each work first *identifies* a structural bottleneck, then *expresses* it as a quantitative bound with a controllable parameter, and finally *constructs* algorithms or options that systematically drive that parameter to tighten the bound, all while preserving the standard PAC guarantees of tabular RL.

3 Random Cluster Growth as One-Dimensional Walk

Explore the Erdős–Rényi graph $G(n, \lambda/n)$ by a breadth-first search (BFS) that starts from a single seed vertex v_0 . At each discrete step $t \geq 0$, the vertex set is partitioned into

$$\begin{aligned} A_t &= \text{“active” frontier (discovered, not yet exposed),} \\ U_t &= \text{“unexplored” vertices (never seen so far),} \\ R_t &= \text{“removed” vertices (all neighbours exposed).} \end{aligned}$$

Write $a_t := |A_t|$ and $r_t := |R_t|$, and encode the process by the *cluster-walk*

$$W_t = a_t - r_t, \quad W_0 = 1. \quad (4)$$

When the first active vertex of step t is exposed it contributes -1 to the frontier size and reveals $\text{Binom}((1 - \delta)n, \lambda/n)$ new vertices, for a bookkeeping constant $\delta > 0$ that discounts already encountered vertices. Hence, the increments of [Equation 4](#) are

$$X_t = W_{t+1} - W_t = -1 + \text{Binom}\left((1 - \delta)n, \frac{\lambda}{n}\right). \quad (5)$$

Link to reinforcement learning. If the state-transition graph of a tabular MDP is regarded as the underlying graph, then

$$\begin{aligned} A_t &\longleftrightarrow \text{known but under-explored states,} \\ U_t &\longleftrightarrow \text{never-visited states,} \\ R_t &\longleftrightarrow \text{fully explored states.} \end{aligned}$$

Consequently, the walk W_t tracks the net size of the exploration frontier for an *uninformed* RL agent. The drift of [Equation 5](#) plays the same quantitative role as the Laplacian spectral gap $\lambda_2(L)/(2n^2)$ in the Broder–Karlín cover-time bound ([Section 4.3](#)).

4 Cluster-Growth Bounds and Their RL Counterpart

4.1 One-Dimensional Encoding of BFS on $G(n, \lambda/n)$

Durrett [[Dur10](#), Sec. 2.3] represents the breadth-first search that explores the connected component containing a root vertex by the walk

$$W_t = |A_t| - |R_t|, \quad W_0 = 1,$$

whose increments are

$$X_t = -1 + \text{Binom}\left((1 - \delta)n, \frac{\lambda}{n}\right). \quad (6)$$

Drift.

$$\mathbb{E}[X_t] = -1 + (1 - \delta)\lambda = \lambda(1 - \delta) - 1 =: \Delta_{\lambda, \delta}. \quad (7)$$

Subcritical if $\lambda < 1$ ($\Delta_{\lambda, \delta} < 0$); *supercritical* if $\lambda > 1$ and $\delta < 1 - \lambda^{-1}$.

Large-deviation tail for extinction. Let $T_0 := \inf\{t : W_t = 0\}$. Cramér-Chernoff gives

$$\Pr_1(T_0 < \infty) \leq \exp(-\theta_\delta W_0), \quad \theta_\delta := \sup_{\theta > 0} \left\{ \theta - \log \mathbb{E}[e^{\theta X_t}] \right\}. \quad (8)$$

For Equation 6, $\mathbb{E}[e^{\theta X_t}] = \exp\left(\frac{\lambda(1-\delta)}{n}(e^\theta - 1) - \theta\right)$, and optimisation yields

$$e^{-\theta_\delta} = e^{-\theta_\delta + \lambda(1-\delta)(e^{\theta_\delta} - 1)}, \quad \theta_\delta \approx \Delta_{\lambda,\delta} \quad (\theta_\delta \ll 1). \quad (9)$$

If $W_0 = \gamma \log n$ with $\gamma > 2/\theta_\delta$ then

$$\Pr(T_0 < \infty) \leq n^{-2}. \quad (10)$$

Hence, the component lives long enough to reach size $(1 - \rho(\lambda))n$, where $\rho(\lambda)$ solves $x = e^{-\lambda(1-x)}$.

4.2 Broder-Karlin Cover-Time Tail

For a lazy uniform random walk on an n -vertex undirected graph with normalised Laplacian L and algebraic connectivity $\lambda_2 := \lambda_2(L)$, Broder and Karlin [BK89] show

$$\Pr[C(G) > t] \leq \exp\left(-\frac{\lambda_2}{2n^2} t\right), \quad t \geq 2n^2 \log n. \quad (11)$$

Integrating the tail,

$$\mathbb{E}[C(G)] \leq \frac{2n^2 \log n}{\lambda_2} (1 + o(1)). \quad (12)$$

4.3 Parameter Identification

$$\boxed{\lambda(1 - \delta) - 1 \longleftrightarrow \frac{\lambda_2}{2n^2}} \quad (13)$$

Thus *Galton-Watson drift* $\Delta_{\lambda,\delta}$ is the RL analogue of the *Laplacian spectral gap*.

4.4 Crossing the Critical Point

Durrett's giant-component condition $\lambda(1 - \delta) > 1$ translates, via Equation 13, into

$$\lambda_2 > \frac{2n^2}{\log n}. \quad (14)$$

If Equation 14 holds, then from Equation 12, we have $\mathbb{E}[C(G)] = O(n \log n)$.

4.5 Effect of *Covering Options*

Now, we see that adding a shortcut edge (u, v) (i.e. a *point-option*) perturbs λ_2 by the first-order eigen-perturbation [GB06]:

$$\lambda_2^{\text{new}} = \lambda_2^{\text{old}} + \frac{(f_v - f_u)^2}{2 R_{\text{eff}}(u, v)} + O((f_v - f_u)^3), \quad (15)$$

where f is the Fiedler vector. Choosing $u = \arg \max f$, $v = \arg \min f$ maximises the numerator, monotonically increasing λ_2 until Equation 14 is satisfied.

4.6 Unified Statement

Theorem 1 (Informal). Breadth-first search on $G(n, \lambda/n)$ and an uninformed random walk on a state graph share the *same* exponential decay rate for extinction/cover time once the identification (Equation 13) is made. Adding long-range edges—whether literal edges or RL point-options—raises that rate above its critical threshold, causing (respectively) a giant graph component or rapid state-space coverage.

4.7 Practical RL Recipe

1. Estimate $\lambda_2(L)$ (e.g. via successor features).
2. If $\lambda_2 \leq 2n^2/\log n$, iteratively add covering options using the Fiedler vector.
3. Re-estimate λ_2 after each insertion; stop once Equation 14 holds.
4. Guaranteed: a uniform random walk now achieves $\mathbb{E}[C(G)] = O(n \log n)$, and any stronger exploration strategy inherits the same order bound.

Dictionary

Durrett (graph)	RL analogue
Active set A_t	Frontier of known-but-under-explored states
Unexplored U_t	Never-visited states
Removed R_t	Fully sampled states
Drift $\lambda(1 - \delta) - 1$	Spectral gap $\lambda_2/(2n^2)$
Extinction event $W_t \rightarrow 0$	Failure to reach sparse reward
Chernoff tail $\exp(-\theta_\delta t)$	Cover-time tail $\exp(-\lambda_2 t/2n^2)$
Giant component	Post-cover-time visited set

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